ALLIANCE UNIVERSITY

Database Management System

Assignment 8

CAB BOOKING SYSTEM

Name	Razin Basha R
Roll No	2511021240012
Course	Database Management System
Programme	BCA AI/ML
Semester	III

Filtering records using WHERE

Query 1: Completed trips longer than 30 km

```
SELECT Booking_ID, Customer_ID, Pickup_Location, Drop_Location, Distance_KM
FROM Booking
WHERE Booking_Status = 'Completed' AND Distance_KM > 30;
```

Output:

Booking_ID	Customer_ID	Pickup_Location	Drop_Location	Distance_KM
B04	CU04	Whitefield	Airport	38
B07	CU02	Airport	Hosur	45
B09	CU04	Airport	Whitefield	36
B11	CU01	Electronic City	Airport	40
B12	CU03	Krishnagiri	Hosur	35

Query 2: Customers living in Bangalore or Chennai (IN)

```
SELECT Customer_ID, Customer_Name, Address
FROM Customer
WHERE Address IN ('Bangalore', 'Chennai');
```

Output:

Customer_ID	Customer_Name	Address
CU01	Rahul	Bangalore
CU02	Arjun	Chennai
CU04	Priya	Bangalore
CU05	Meena	Chennai
CU06	Vikram	Bangalore

Query 3: Bookings between two dates

```
SELECT Booking_ID, Booking_Date, Pickup_Location, Drop_Location
FROM Booking
WHERE Booking_Date BETWEEN '2026-09-10' AND '2026-09-15';
```

Output:

Booking_ID	Booking_Date	Pickup_Location	Drop_Location
B05	2026-09-10	MG Road	Electronic City
B06	2026-09-11	Railway Station	T Nagar
B07	2026-09-12	Airport	Hosur
B08	2026-09-13	Koramangala	Indiranagar
B09	2026-09-14	Airport	Whitefield
B10	2026-09-15	Mall	College

Query 4: Cars whose type starts with S (LIKE)

```
SELECT Car_ID, Car_Model, Car_Type, Fare_Per_KM
FROM Car
WHERE Car_Type LIKE 'S%';
```

Output:

Car_ID	Car_Model	Car_Type	Fare_Per_KM
C02	City	Sedan	15.00

Car_ID	Car_Model	Car_Type	Fare_Per_KM
C03	Creta	SUV	20.00
C04	Innova	SUV	22.00
C05	Dzire	Sedan	14.00

Sorting records using ORDER BY

Query 5: The five longest trips, longest first

```
SELECT Booking_ID, Pickup_Location, Drop_Location, Distance_KM
FROM Booking
ORDER BY Distance_KM DESC, Booking_ID
LIMIT 5;
```

Output:

Booking_ID	Pickup_Location	Drop_Location	Distance_KM
B07	Airport	Hosur	45
B11	Electronic City	Airport	40
B04	Whitefield	Airport	38
B09	Airport	Whitefield	36
B12	Krishnagiri	Hosur	35

Query 6: Cars sorted by fare, cheapest first

```
SELECT Car_ID, Car_Model, Car_Type, Fare_Per_KM
FROM Car
ORDER BY Fare_Per_KM ASC;
```

Output:

Car_ID	Car_Model	Car_Type	Fare_Per_KM
C06	Alto	Hatchback	10.00
C07	Baleno	Hatchback	11.00
C01	Swift	Hatchback	12.00
C05	Dzire	Sedan	14.00
C02	City	Sedan	15.00
C03	Creta	SUV	20.00
C04	Innova	SUV	22.00

Aggregate functions

Query 7: COUNT of bookings

```
SELECT COUNT(*) AS Total_Bookings FROM Booking;
```

Output:

Total_Bookings
16

Query 8: SUM, AVG, MIN and MAX of the money received

```
SELECT COUNT(*)            AS Paid_Payments,
       SUM(Amount)          AS Total_Amount,
       ROUND(AVG(Amount),2) AS Avg_Amount,
       MIN(Amount)          AS Min_Amount,
       MAX(Amount)          AS Max_Amount
FROM Payment
```

```
WHERE Payment_Status = 'Paid';
```

Output:

Paid_Payments	Total_Amount	Avg_Amount	Min_Amount	Max_Amount
11	4522.00	411.09	50.00	900.00

Query 9: Distance statistics of all bookings

```
SELECT COUNT(*)                  AS Trips,
       SUM(Distance_KM)           AS Total_KM,
       ROUND(AVG(Distance_KM),2) AS Avg_KM,
       MIN(Distance_KM)           AS Shortest_KM,
       MAX(Distance_KM)           AS Longest_KM
FROM Booking;
```

Output:

Trips	Total_KM	Avg_KM	Shortest_KM	Longest_KM
16	393	24.56	5	45

Grouping records using GROUP BY

Query 10: Number of bookings in each status

```
SELECT Booking_Status, COUNT(*) AS Total_Bookings
FROM Booking
GROUP BY Booking_Status
ORDER BY Total_Bookings DESC;
```

Output:

Booking_Status	Total_Bookings
Completed	11
Confirmed	3
Cancelled	2

Query 11: Payments received by each payment method

```
SELECT Payment_Method,
       COUNT(*)    AS Payments,
       SUM(Amount) AS Total_Amount
FROM Payment
WHERE Payment_Status = 'Paid'
GROUP BY Payment_Method
ORDER BY Total_Amount DESC;
```

Output:

Payment_Method	Payments	Total_Amount
Card	4	2796.00
UPI	4	1530.00
Cash	3	196.00

Query 12: Fare statistics for each car type

```
SELECT Car_Type,
       COUNT(*)                  AS Cars,
       ROUND(AVG(Fare_Per_KM),2) AS Avg_Fare,
       MIN(Fare_Per_KM)          AS Min_Fare,
       MAX(Fare_Per_KM)          AS Max_Fare
FROM Car
GROUP BY Car_Type
ORDER BY Avg_Fare;
```

Output:

Car_Type	Cars	Avg_Fare	Min_Fare	Max_Fare
Hatchback	3	11.00	10.00	12.00
Sedan	2	14.50	14.00	15.00
SUV	2	21.00	20.00	22.00

8.5 Filtering grouped results using HAVING

Query 13: Customers who made more than one booking

```
SELECT Customer_ID,
       COUNT(*)          AS Total_Bookings,
       SUM(Distance_KM) AS Total_KM
FROM Booking
GROUP BY Customer_ID
HAVING COUNT(*) > 1
ORDER BY Total_Bookings DESC, Customer_ID;
```

Output:

Customer_ID	Total_Bookings	Total_KM
CU01	3	87
CU02	3	62
CU03	2	65
CU04	2	74
CU05	2	29
CU06	2	42
CU07	2	34

Query 14: Cars that travelled more than 60 km in total (cancelled trips ignored)

```
SELECT Car_ID,
       COUNT(*)          AS Trips,
       SUM(Distance_KM) AS Total_KM
FROM Booking
WHERE Booking_Status <> 'Cancelled'
GROUP BY Car_ID
HAVING SUM(Distance_KM) > 60
ORDER BY Total_KM DESC;
```

Output:

Car_ID	Trips	Total_KM
C03	3	108
C04	3	106
C01	2	61

Manipulating data (UPDATE)

Query 15: A busy driver becomes available again

```
UPDATE Driver SET Availability = 'Available' WHERE Driver_ID = 'D06';
```

```
SELECT Driver_ID, Driver_Name, Availability FROM Driver ORDER BY Driver_ID;
```

Output:

Driver_ID	Driver_Name	Availability
D01	Ravi	Available
D02	Kumar	Busy
D03	Suresh	Available
D04	Mohan	Available
D05	Prakash	Available
D06	Selvam	Available